

REV

A

DESCRIPTION

First issue, for quotation and engineering review

DATE

2026-10-05

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FIREFLY FRAMES

Fabrication drawings for the two block frames, Rib and Hoop, and the parts they share

SHEETS

SHEET	TITLE	PART NO.	SET
G-1	Drawing set index and general notes		Index
R-0	Rib frame assembly	FF-R00	Rib
R-1	Rib side walls, right and left	FF-R01, FF-R02	Rib
R-2	Rib side wall, flat pattern	FF-R01, FF-R02	Rib
R-3	Rib top and bottom walls	FF-R03, FF-R04	Rib
R-4	Rib seam rail	FF-R05	Rib
R-5	Rib hub hanger	FF-R06	Rib
R-6	Rib corner and tab details	FF-R01 to FF-R04	Rib
H-0	Hoop frame assembly	FF-H00	Hoop
H-1	Hoop strip: profile, corners and weld	FF-H01	Hoop
H-2	Hoop strip, flat pattern	FF-H01	Hoop
H-3	Hoop strip, punch details	FF-H01	Hoop
H-4	Hoop seam rail	FF-H02	Hoop
H-5	Hoop hub mount	FF-H03, FF-H04, FF-H05	Hoop
C-1	Gravity lock, panel to frame	J groove	Common
C-2	Block joints	Joint features	Common
C-3	Base rail	FF-C03, FF-C04	Common
C-4	Mounting: hole grid, angle and tube clips	FF-C05	Common
C-5	Locating pins and hardware	FF-C01, FF-C02	Common

THE FOUR BLOCKS, NAMED HEIGHT x WIDTH IN FEET

BLOCK	OUTSIDE W x H x D	PANELS	JOINT BOLTS	RIB FRAME, kg	HOOP FRAME, kg
3x3	914.4 x 914.4 x 152.4	9	7	5.4	12.7
3x2	609.6 x 914.4 x 152.4	6	6	4.1	9.5
2x3	914.4 x 609.6 x 152.4	6	5	4.2	10.2
2x2	609.6 x 609.6 x 152.4	4	4	3.2	7.5

1. The 3x2 is 3 high and 2 wide, a part of its own, never a 2x3 turned on end. Every block hangs one way only.

2. Frame masses are approximate, bare frame before powder coat, without panels or hub.

FROM THE FRONT, 1:40. THE HUB SITS BEHIND THE SHADED PANEL.

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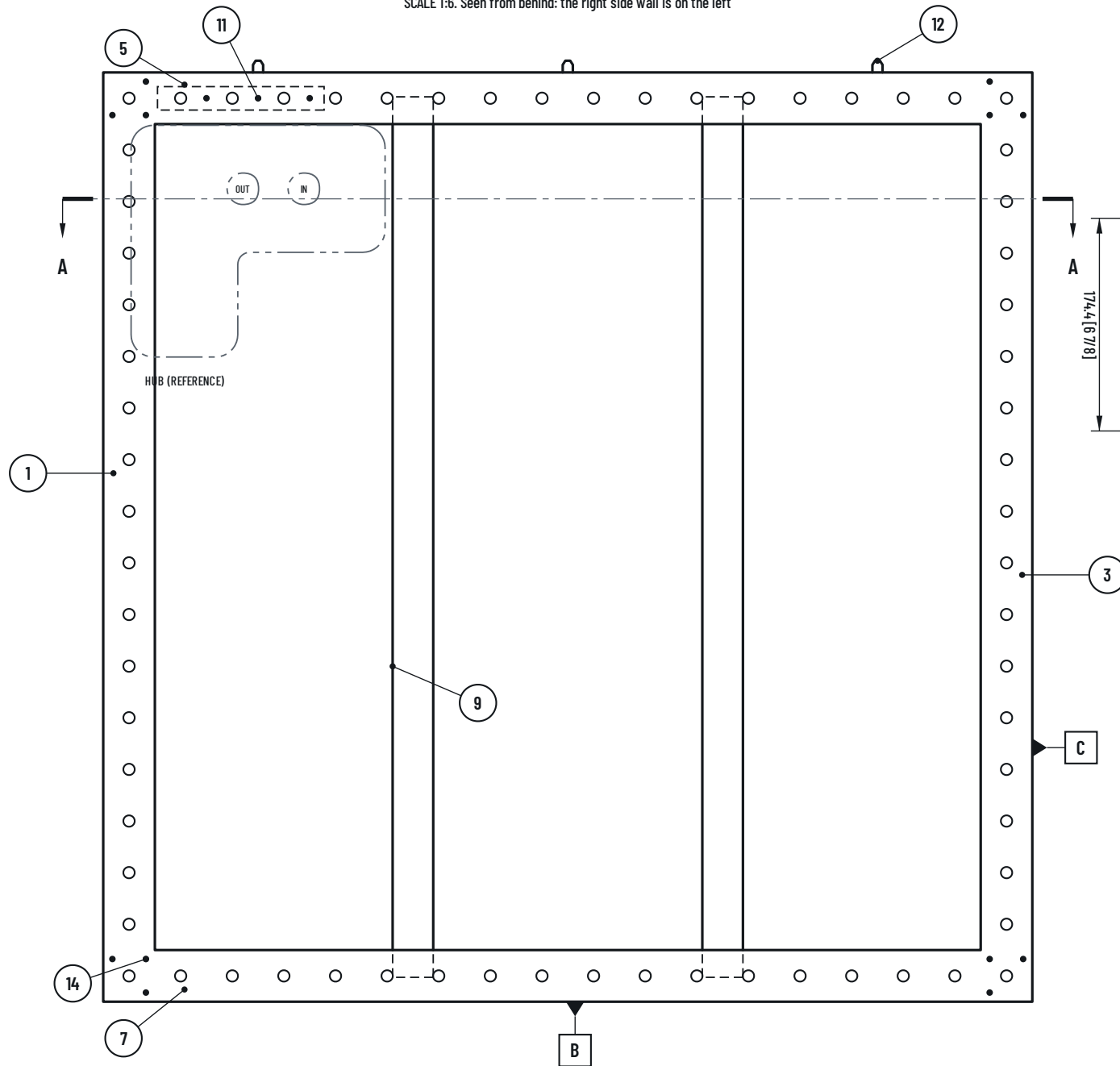
2x3

2x2

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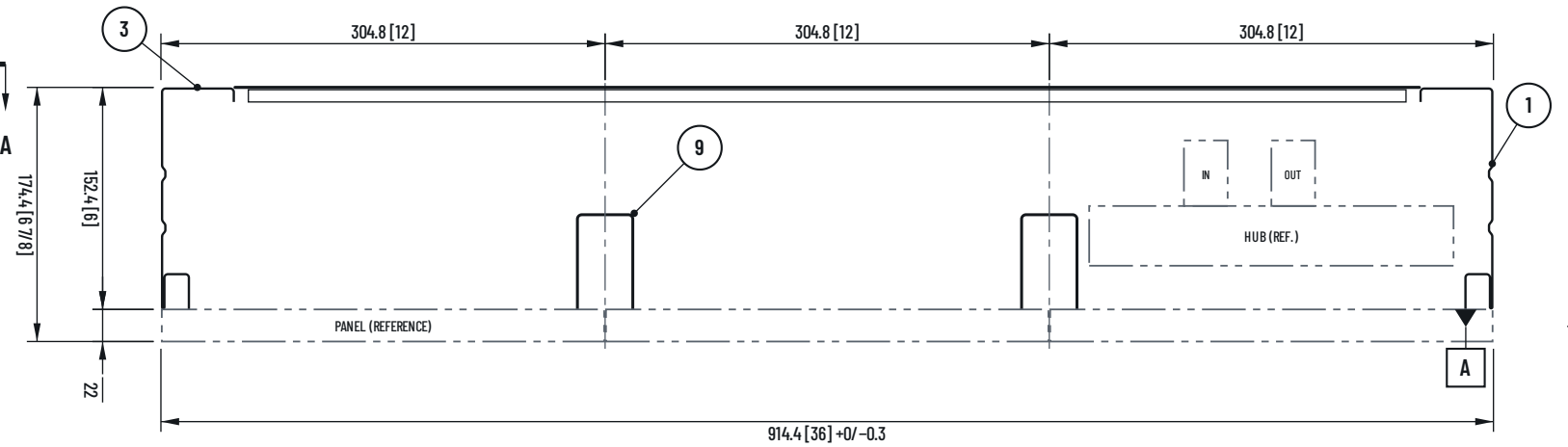
REAR VIEW, 3x3 BLOCK

SCALE 1:6. Seen from behind: the right side wall is on the left



SECTION A-A

SCALE 1:5. Looking down at 790 above the bottom face, front edge down



ASSEMBLY NOTES

1. Assemble in a fixture located on datums A (front plane), B (bottom face) and C (left face, seen from the front). Outside size $+0/-0.3$ each way, diagonals equal within 0.5, back flange flat within 0.5.
2. Order: side wall tabs inside the end walls, corner rivets through the overlapping flanges, seam rails, then the hanger. Parts arrive powder coated with nuts and pins already pressed in.
3. Every rivet is flush on an outside face (top, bottom or back). The side walls' outer faces carry no fastener: they hold the logo field and the side joints.
4. Joint pins, nuts and sockets: sheet C-2. Back flange holes: sheet C-4. Gravity lock grooves: sheet C-1.
5. The hub is its own assembly, shown for reference. It hangs on item 1l and locks with its own spring pin.
6. Blocks are named height x width in feet: 3x3, 3x2 (3 high, 2 wide), 2x3 (2 high, 3 wide) and 2x2.

BILL OF MATERIALS, PER BLOCK. QUANTITY FOR EACH BLOCK SIZE (HEIGHT x WIDTH, FT).

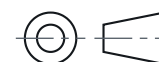
ITEM	PART NO.	DESCRIPTION	MATERIAL	3x3	3x2	2x3	2x2
1	FF-R01-3	Side wall, right hand, 3 ft, PEM nuts pressed in	5052-H32, 1.6	1	1		
2	FF-R01-2	Side wall, right hand, 2 ft, PEM nuts pressed in	5052-H32, 1.6			1	1
3	FF-R02-3	Side wall, left hand, 3 ft	5052-H32, 1.6	1	1		
4	FF-R02-2	Side wall, left hand, 2 ft	5052-H32, 1.6			1	1
5	FF-R03-3	Top wall, 3 ft wide, pins and PEM nuts pressed in	5052-H32, 1.6	1		1	
6	FF-R03-2	Top wall, 2 ft wide, pins and PEM nuts pressed in	5052-H32, 1.6		1		1
7	FF-R04-3	Bottom wall, 3 ft wide, pin sockets	5052-H32, 1.6	1		1	
8	FF-R04-2	Bottom wall, 2 ft wide, pin sockets	5052-H32, 1.6		1		1
9	FF-R05-3	Seam rail, channel, 3 ft	5052-H32, 2.0	2	1		
10	FF-R05-2	Seam rail, channel, 2 ft	5052-H32, 2.0			2	1
11	FF-R06	Hub hanger	5052-H32, 2.0	1	1	1	1
12	FF-C01	Locating pin, press fit, in item 5 or 6 (sheet C-5)	303 stainless	3	2	3	2
13	CLS-M8-1	PEM self-clinching nut M8, in items 1, 2, 5, 6	303 stainless	7	6	5	4
14	RIV-32-CSK	Blind rivet Ø3.2, closed end, 100° countersunk	Al body, A2 mandrel	35	31	35	31
15	ISO 4162	Hex flange bolt M8 x 16, 8.8, ships loose for the joints (sheet C-2)	Steel, zinc flake	7	6	5	4

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS

PROJECT FIREFLY

FRAME: RIB



THIRD ANGLE

DO NOT SCALE THE DRAWING

Preliminary: for quote and review

	TITLE
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RIB FRAME ASSEMBLY

3x3 shown, every block size in the bill of materials

PART NO.
FF-R00-33 / -32 / -23 / -22

QTY PER BLOCK	1
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MATERIAL	See bill of materials
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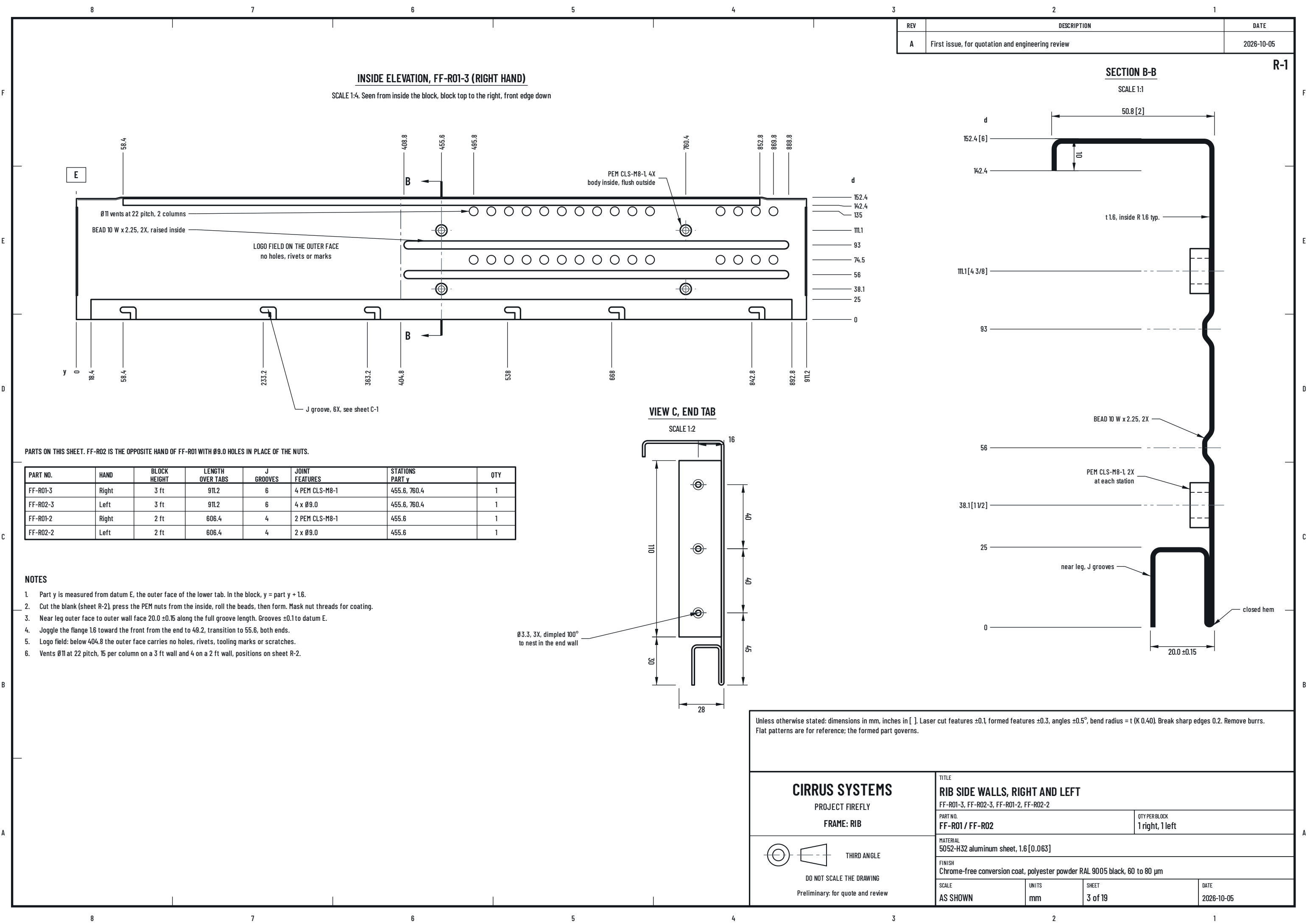
FINISH
 Chrome-free conversion coat, polyester powder RAL 9005 black, 60 to 80 µm, parts coated before riveting

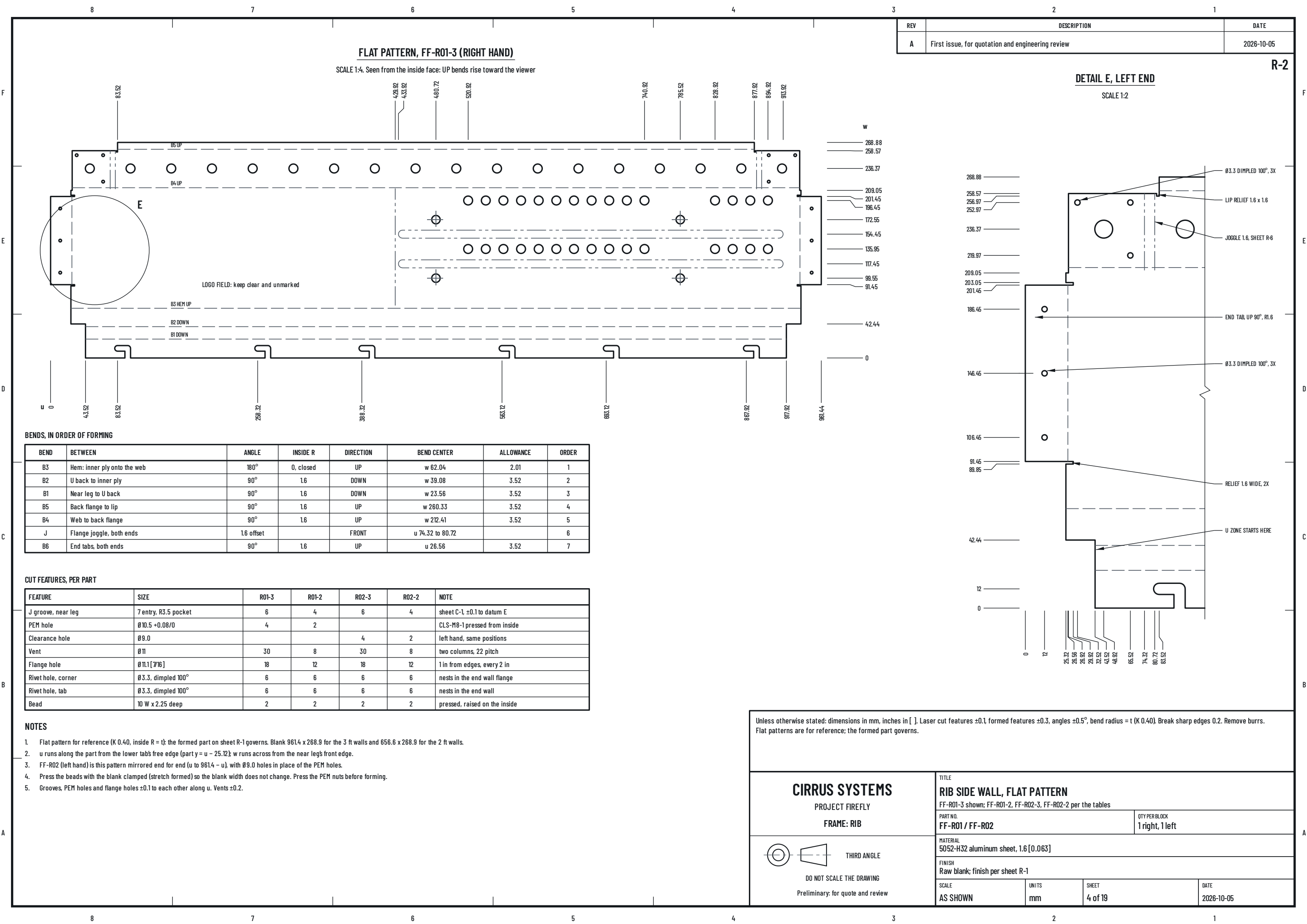
SCALE
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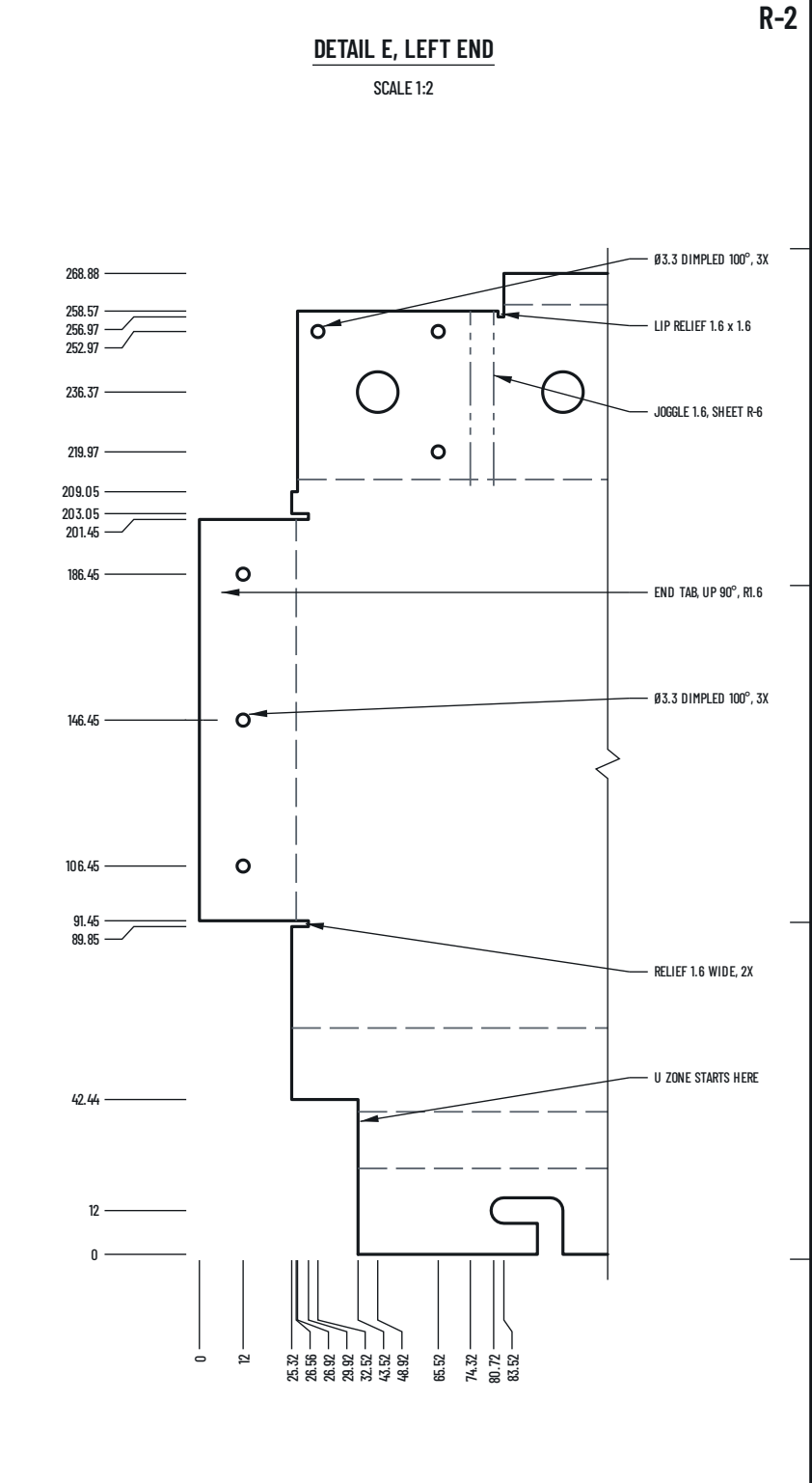
SHEET
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BENDS, IN ORDER OF FORMING

BEND	BETWEEN	ANGLE	INSIDE R	DIRECTION	BEND CENTER	ALLOWANCE	ORDER
B3	Hem: inner ply onto the web	180°	0, closed	UP	w 62.04	2.01	1
B2	U back to inner ply	90°	1.6	DOWN	w 39.08	3.52	2
B1	Near leg to U back	90°	1.6	DOWN	w 23.56	3.52	3
B5	Back flange to lip	90°	1.6	UP	w 260.33	3.52	4
B4	Web to back flange	90°	1.6	UP	w 212.41	3.52	5
J	Flange joggle, both ends	1.6 offset		FRONT	u 74.32 to 80.72		6
B6	End tabs, both ends	90°	1.6	UP	u 26.56	3.52	7

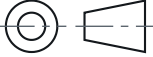
CUT FEATURES, PER PART

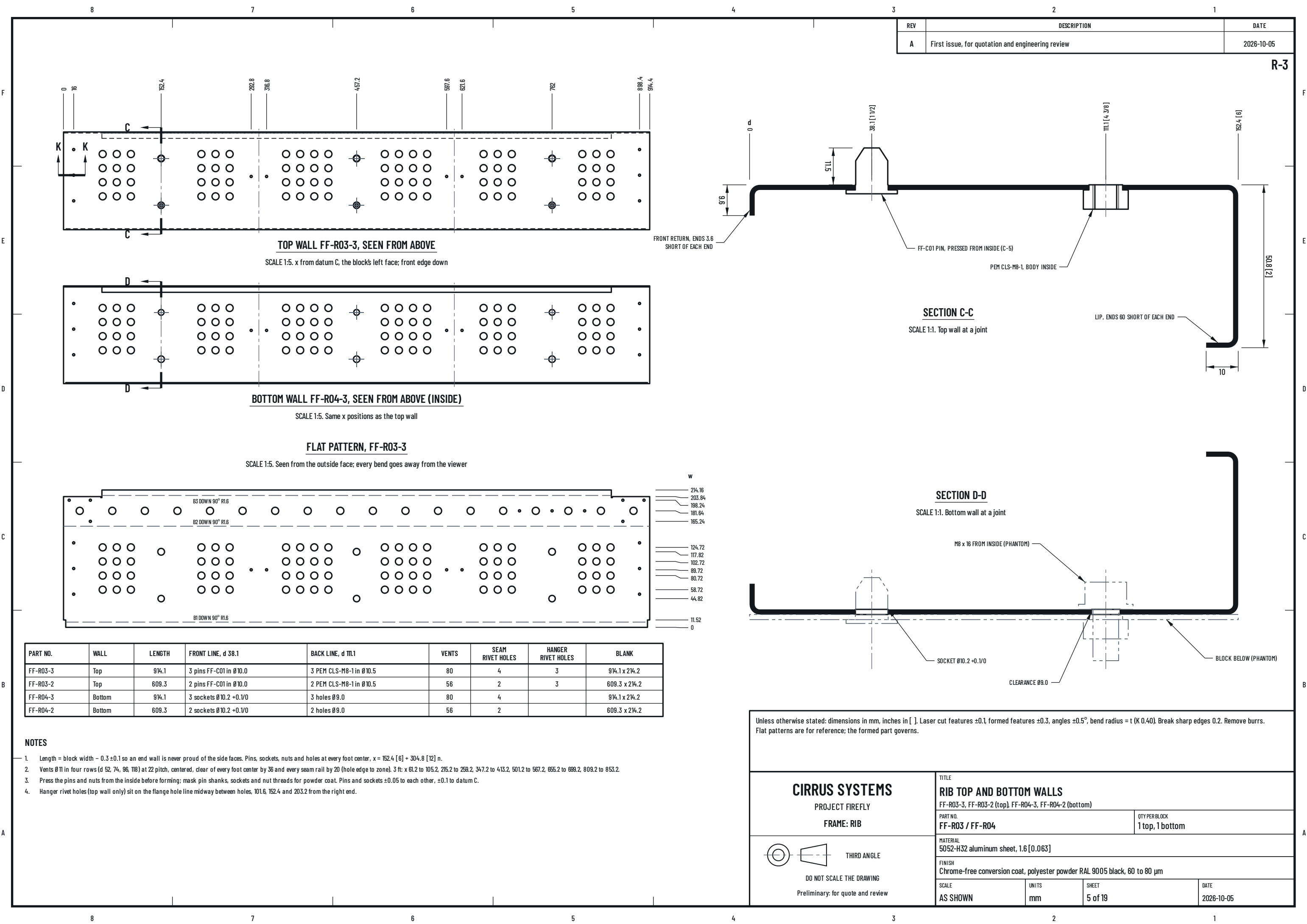
FEATURE	SIZE	R01-3	R01-2	R02-3	R02-2	NOTE
J groove, near leg	7 entry, R3.5 pocket	6	4	6	4	sheet C-1, ±0.1 to datum E
PEM hole	Ø10.5 +0.08/0	4	2			CLS-M8-1 pressed from inside
Clearance hole	Ø9.0			4	2	left hand, same positions
Vent	Ø11	30	8	30	8	two columns, 22 pitch
Flange hole	Ø11.1 [7/16]	18	12	18	12	1 in from edges, every 2 in
Rivet hole, corner	Ø3.3, dimpled 100°	6	6	6	6	nests in the end wall flange
Rivet hole, tab	Ø3.3, dimpled 100°	6	6	6	6	nests in the end wall
Bead	10 W x 2.25 deep	2	2	2	2	pressed, raised on the inside

NOTES

- Flat pattern for reference (K 0.40, inside R = t): the formed part on sheet R-1 governs. Blank 961.4 x 268.9 for the 3 ft walls and 656.6 x 268.9 for the 2 ft walls.
- u runs along the part from the lower tab's free edge (part y = u - 25.12); w runs across from the near leg's front edge.
- FF-R02 (left hand) is this pattern mirrored end for end (u to 961.4 - u), with Ø9.0 holes in place of the PEM holes.
- Press the beads with the blank clamped (stretch formed) so the blank width does not change. Press the PEM nuts before forming.
- Grooves, PEM holes and flange holes ±0.1 to each other along u. Vents ±0.2.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ±0.1, formed features ±0.3, angles ±0.5°, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS PROJECT FIREFLY FRAME: RIB THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	TITLE RIB SIDE WALL, FLAT PATTERN FF-R01-3 shown; FF-R01-2, FF-R02-3, FF-R02-2 per the tables	
	PART NO. FF-R01 / FF-R02	QTY PER BLOCK 1 right, 1 left
	MATERIAL 5052-H32 aluminum sheet, 1.6 [0.063]	
	FINISH Raw blank; finish per sheet R-1	
	SCALE AS SHOWN	UNITS mm
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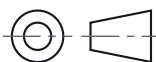


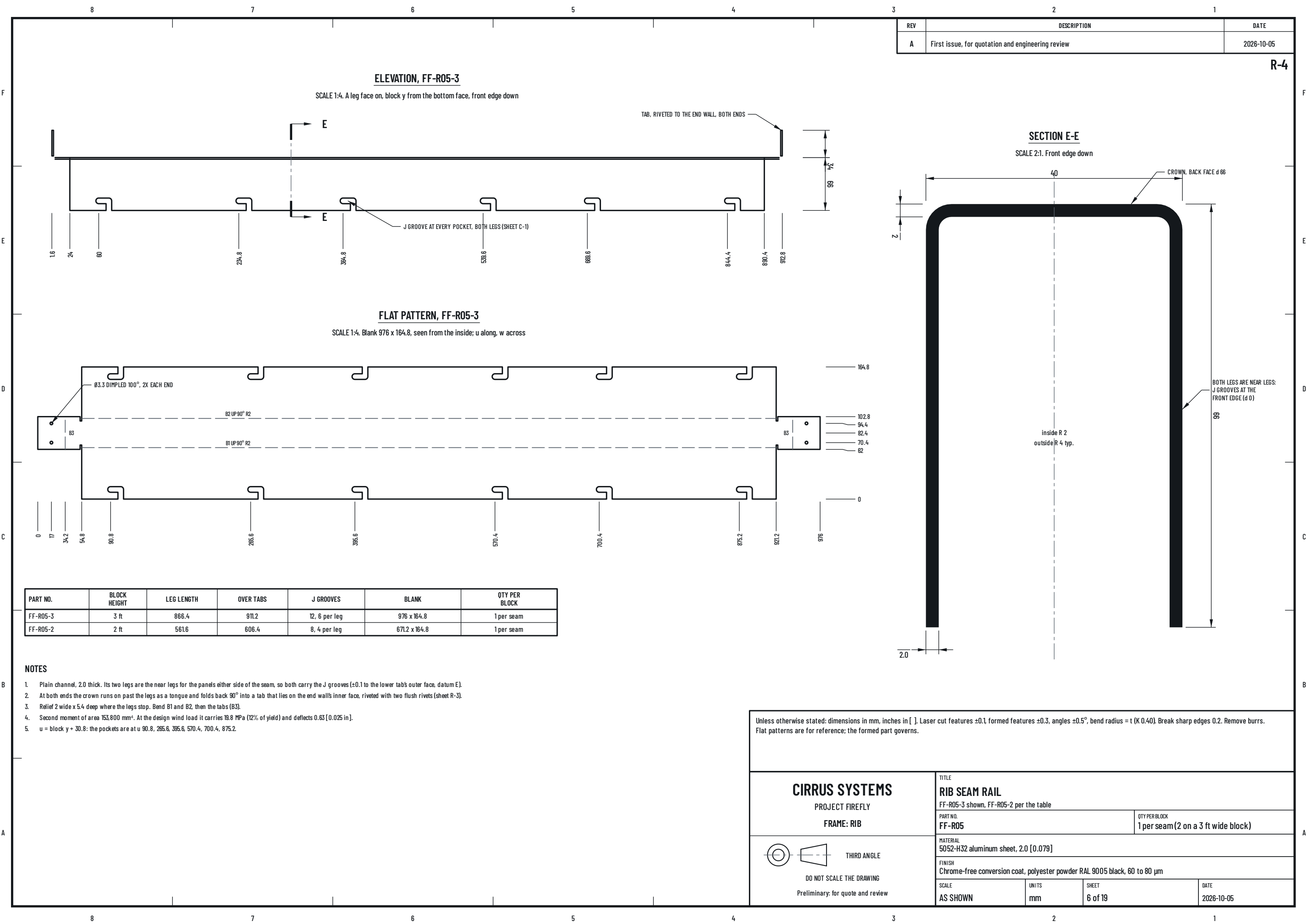
PART NO.	WALL	LENGTH	FRONT LINE, d 38.1	BACK LINE, d 111.1	VENTS	SEAM RIVET HOLES	HANGER RIVET HOLES	BLANK
FF-R03-3	Top	914.1	3 pins FF-C01 in Ø10.0	3 PEM CLS-M8-1 in Ø10.5	80	4	3	914.1 x 214.2
FF-R03-2	Top	609.3	2 pins FF-C01 in Ø10.0	2 PEM CLS-M8-1 in Ø10.5	56	2	3	609.3 x 214.2
FF-R04-3	Bottom	914.1	3 sockets Ø10.2 ±0.1/0	3 holes Ø9.0	80	4		914.1 x 214.2
FF-R04-2	Bottom	609.3	2 sockets Ø10.2 ±0.1/0	2 holes Ø9.0	56	2		609.3 x 214.2

NOTES

- Length = block width - 0.3±0.1 so an end wall is never proud of the side faces. Pins, sockets, nuts and holes at every foot center, x = 152.4 [6] + 304.8 [12] n.
- Vents Ø11 in four rows (d 52, 74, 96, 118) at 22 pitch, centered, clear of every foot center by 36 and every seam rail by 20 (hole edge to zone). 3 ft: x 61.2 to 105.2, 215.2 to 258.2, 347.2 to 413.2, 501.2 to 567.2, 655.2 to 698.2, 809.2 to 853.2.
- Press the pins and nuts from the inside before forming; mask pin shanks, sockets and nut threads for powder coat. Pins and sockets ±0.05 to each other, ±0.1 to datum C.
- Hanger rivet holes (top wall only) sit on the flange hole line midway between holes, 101.6, 152.4 and 203.2 from the right end.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ±0.1, formed features ±0.3, angles ±0.5°, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS PROJECT FIREFLY FRAME: RIB THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	TITLE RIB TOP AND BOTTOM WALLS FF-R03-3, FF-R03-2 (top), FF-R04-3, FF-R04-2 (bottom)			
	PART NO. FF-R03 / FF-R04		QTY PER BLOCK 1 top, 1 bottom	
	MATERIAL 5052-H32 aluminum sheet, 1.6 [0.063]			
	FINISH Chrome-free conversion coat, polyester powder RAL 9005 black, 60 to 80 µm			
	SCALE AS SHOWN	UNITS mm	SHEET 5 of 19	DATE 2026-10-05



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R-4

F

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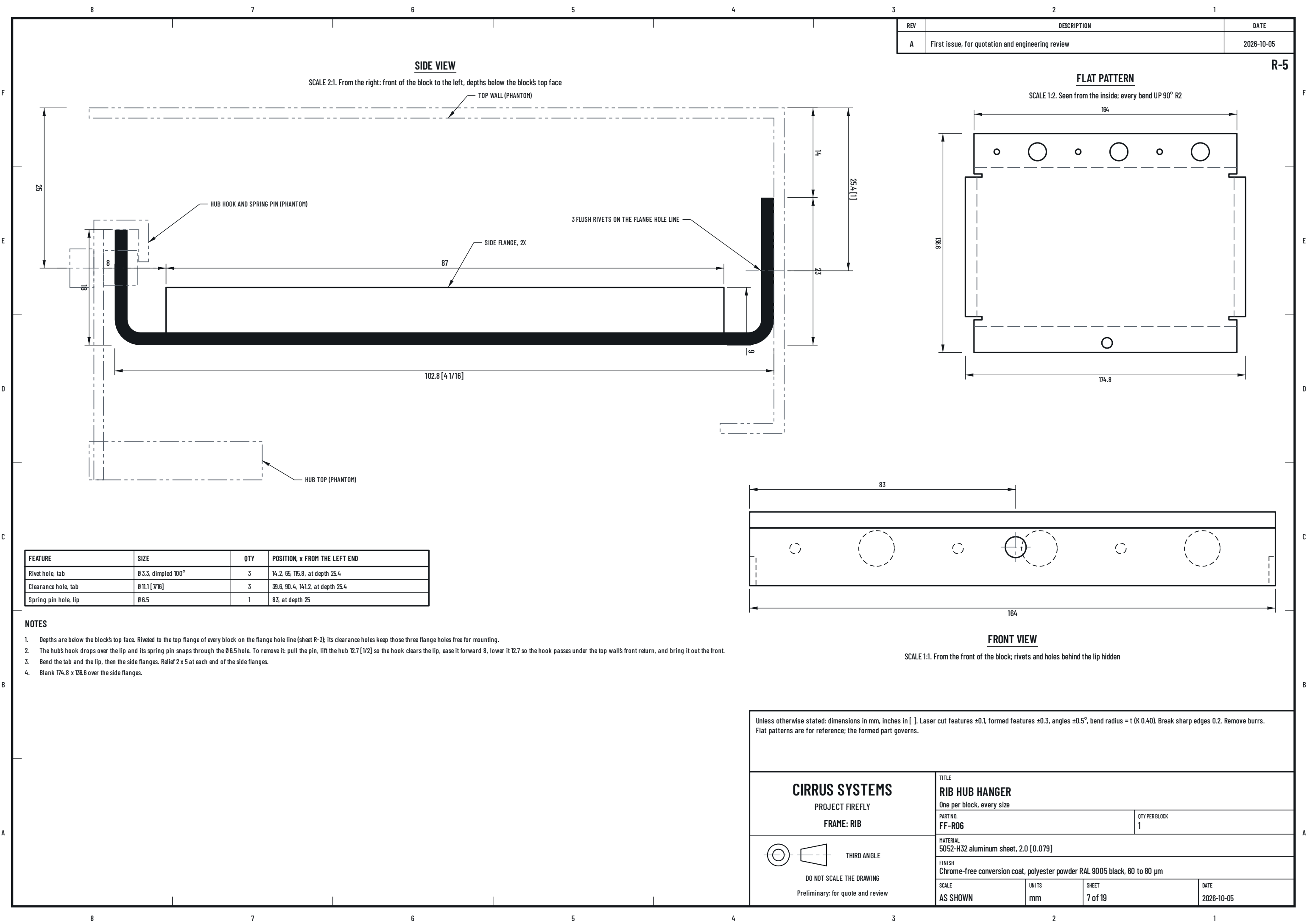
D

C

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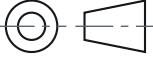
Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.			
CIRRUS SYSTEMS PROJECT FIREFLY FRAME: RIB		TITLE RIB SEAM RAIL FF-R05-3 shown, FF-R05-2 per the table	
		PART NO. FF-R05	QTY PER BLOCK 1 per seam (2 on a 3 ft wide block)
MATERIAL 5052-H32 aluminum sheet, 2.0 [0.079]			
FINISH Chrome-free conversion coat, polyester powder RAL 9005 black, 60 to 80 μm			
SCALE AS SHOWN	UNITS mm	SHEET 6 of 19	DATE 2026-10-05

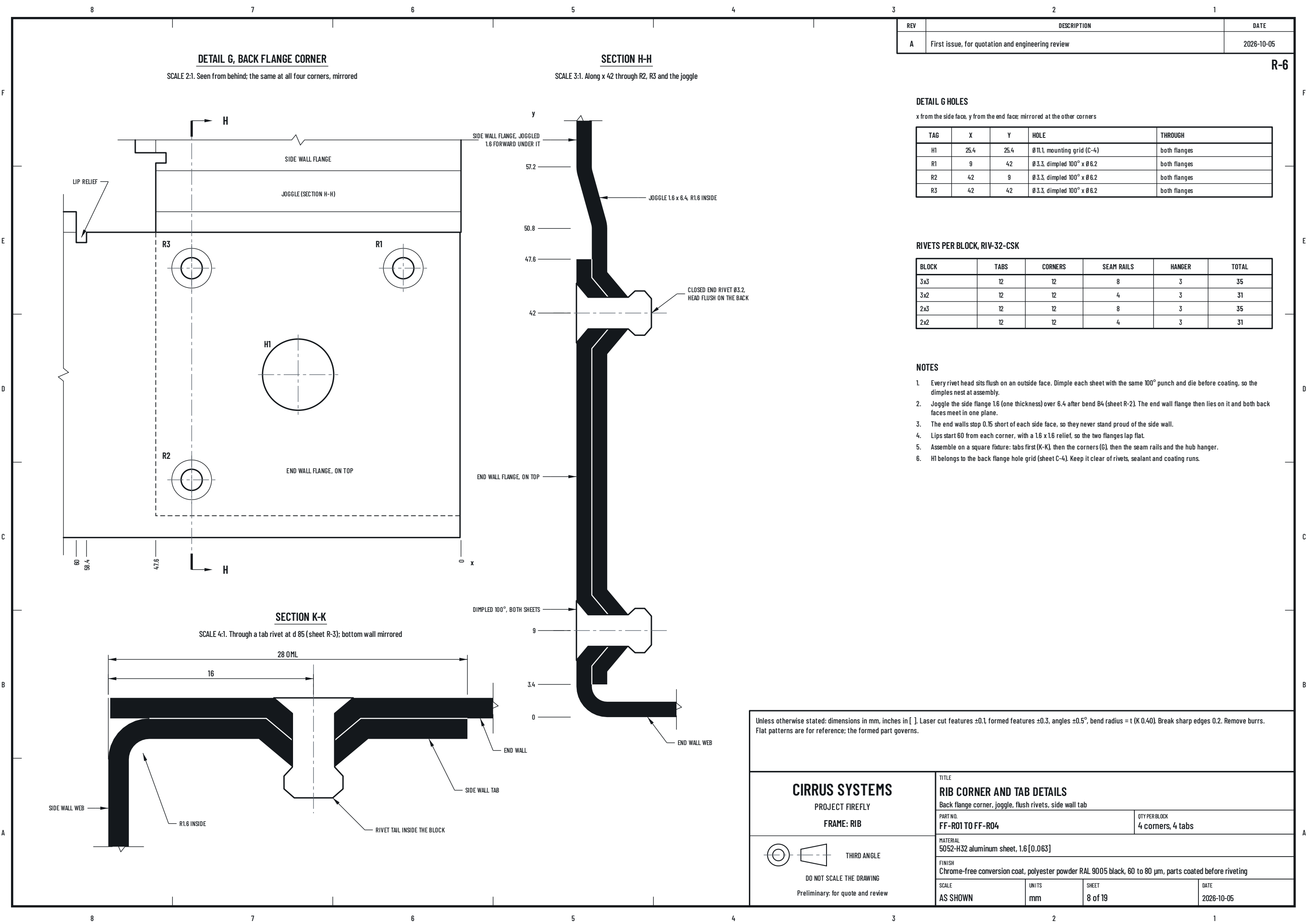


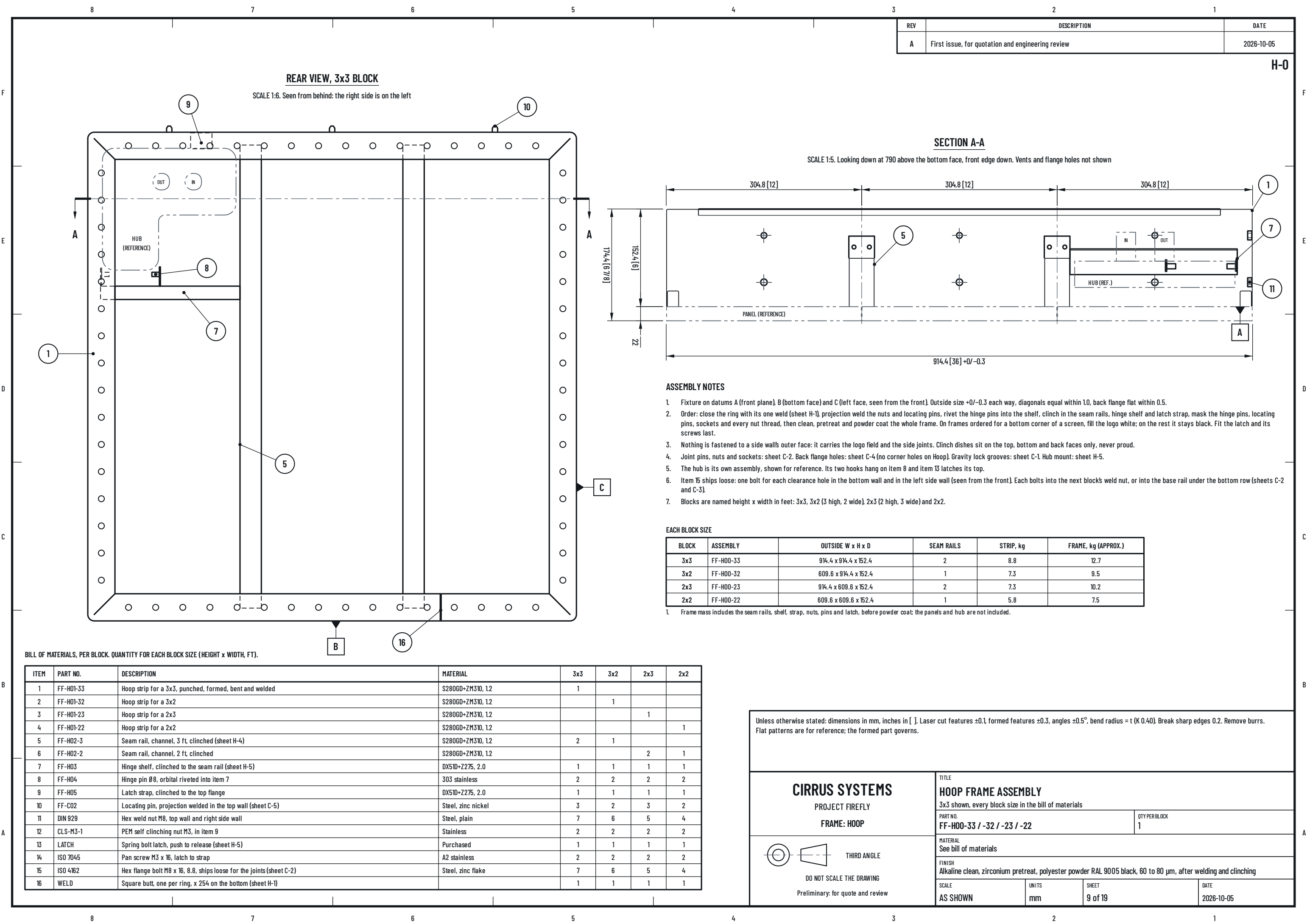
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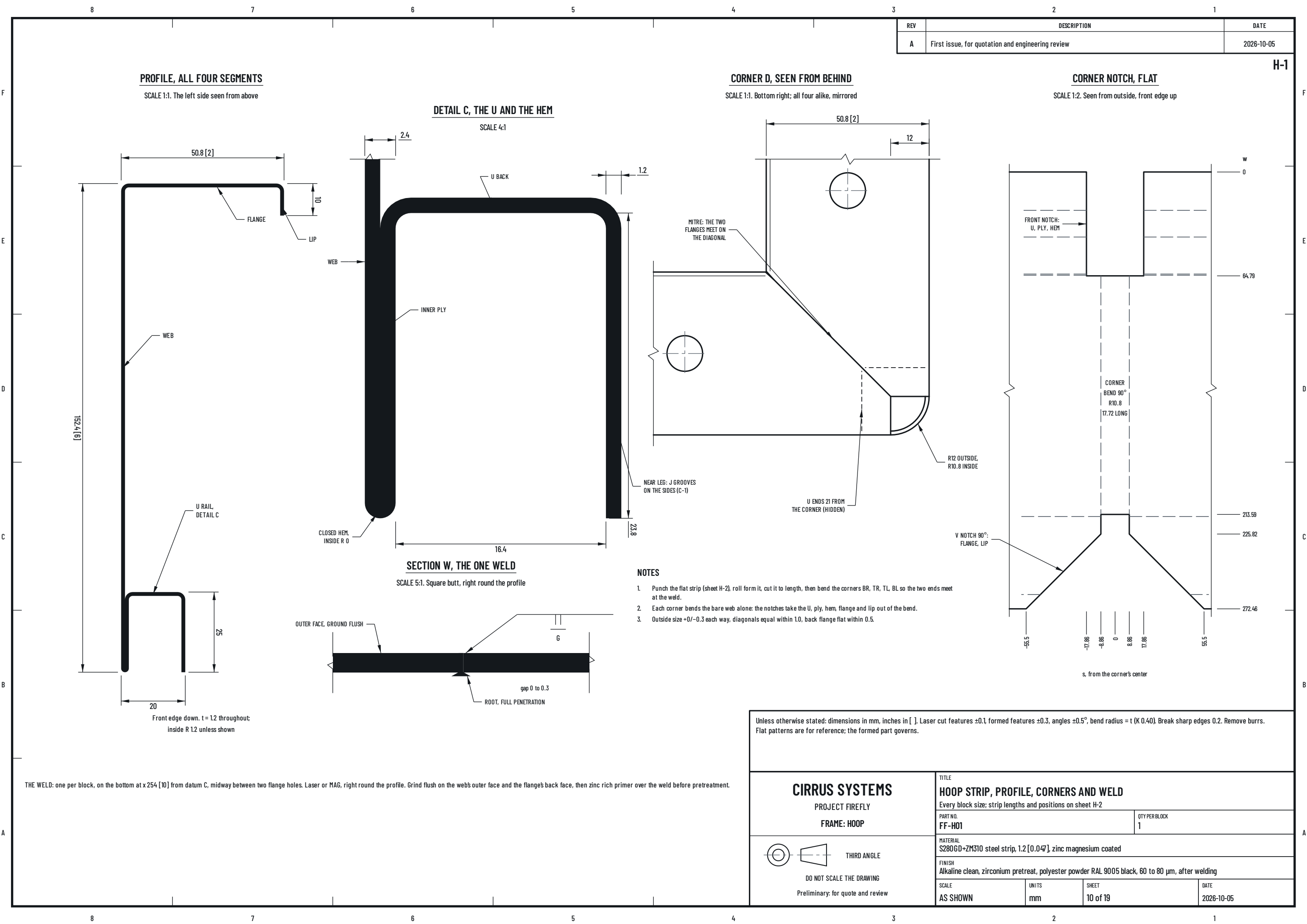
R-5

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ±0.1, formed features ±0.3, angles ±0.5°, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS PROJECT FIREFLY FRAME: RIB THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	TITLE RIB HUB HANGER One per block, every size		
	PART NO. FF-R06		QTY PER BLOCK 1
	MATERIAL 5052-H32 aluminum sheet, 2.0 [0.079]		
	FINISH Chrome-free conversion coat, polyester powder RAL 9005 black, 60 to 80 µm		
	SCALE AS SHOWN	UNITS mm	SHEET 7 of 19 DATE 2026-10-05







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H-1

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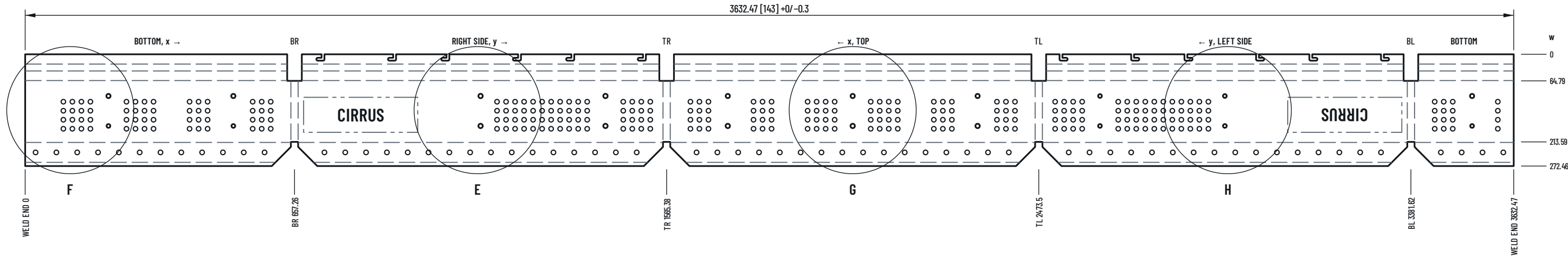
Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.			
CIRRUS SYSTEMS PROJECT FIREFLY FRAME: HOOP		TITLE HOOP STRIP, PROFILE, CORNERS AND WELD Every block size; strip lengths and positions on sheet H-2	
		PART NO. FF-H01	QTY PER BLOCK 1
MATERIAL S280GD+ZM310 steel strip, 1.2 [0.047], zinc magnesium coated			
FINISH Alkaline clean, zirconium pretreat, polyester powder RAL 9005 black, 60 to 80 μm , after welding			
SCALE AS SHOWN	UNITS mm	SHEET 10 of 19	DATE 2026-10-05

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FLAT STRIP, FF-H01-33 (3x3 BLOCK)

SCALE 1:10. Seen from outside, front edge up, s from the weld end to the right

H-2



Details E to H on sheet H-3; F shows both ends butted at the weld. Corner bends run 8.86 either side of each center.

STRIP FOR EACH BLOCK SIZE (HEIGHT x WIDTH, FT)

PART NO.	BLOCK	OUTSIDE W x H	LENGTH L	s, BR	s, TR	s, TL	s, BL	MASS, kg
FF-H01-33	3x3	914.4 x 914.4	3632.47	657.26	1565.38	2473.5	3381.62	8.8
FF-H01-32	3x2	609.6 x 914.4	3022.87	352.46	1260.58	1863.9	2772.02	7.3
FF-H01-23	2x3	914.4 x 609.6	3022.87	657.26	1260.58	2168.7	2772.02	7.3
FF-H01-22	2x2	609.6 x 609.6	2413.27	352.46	955.78	1559.1	2162.42	5.8

CUT FEATURES AND HARDWARE PER STRIP

PART NO.	JOINT HOLES	VENTS	FLANGE HOLES	J GROOVES	CORNER NOTCHES	LOGO DEBOSS	WELD NUTS DIN 929	PINS FF-C02
FF-H01-33	20	272	64	12	8	2	7	3
FF-H01-32	16	224	52	12	8	2	6	2
FF-H01-23	16	184	52	8	8	2	5	3
FF-H01-22	12	136	40	8	8	2	4	2

EVERY CUT FEATURE, ALONG ITS SEGMENT IN BLOCK COORDINATES

FEATURE	SEGMENT	ACROSS, w	SIZE	ALONG, 3 FT SEGMENT	ALONG, 2 FT SEGMENT
Pin socket	Bottom	101.69 (d 38.1)	Ø10.2 +0.1/0	x 152.4, 457.2, 762	x 152.4, 457.2
Bolt clearance	Bottom	174.69 (d 111.1)	Ø9.0	x 152.4, 457.2, 762	x 152.4, 457.2
Pin FF-C02 seat	Top	101.69 (d 38.1)	Ø10.2 +0.1/0	x 152.4, 457.2, 762	x 152.4, 457.2
Weld nut DIN 929 M8	Top	174.69 (d 111.1)	Ø10.6 +0.1/0	x 152.4, 457.2, 762	x 152.4, 457.2
Weld nut DIN 929 M8	Right side	101.69 (d 38.1), 174.69	Ø10.6 +0.1/0	y 457.2, 762	y 457.2
Bolt clearance	Left side	101.69 (d 38.1), 174.69	Ø9.0	y 457.2, 762	y 457.2
J groove, pocket	Both sides	0 to 15.5	7 wide, R3.5	y 60, 234.8, 364.8, 539.6, 669.6, 844.4	y 60, 234.8, 364.8, 539.6
Vent, 4 rows	Bottom	115.59, 137.59, 159.59, 181.59	Ø11, 22 pitch	x 61.2 (3), 215.2, 347.2 (4), 501.2 (4), 655.2 (3), 809.2 (3)	x 51.8 (3), 205.8 (2), 337.8 (4), 513.8 (3)
Vent, 4 rows	Top	the same	Ø11, 22 pitch	x 61.2 (3), 215.2 (3), 347.2 (4), 501.2 (4), 655.2 (3), 809.2 (3)	x 51.8 (3), 205.8 (4), 337.8 (4), 513.8 (3)
Vent, 4 rows	Both sides	the same	Ø11, 22 pitch	y 497.4 (11), 805.4 (4)	y 499 (4)
Flange hole	All four	239.22 (q 25.4)	Ø11.1 50.8 pitch	from 76.2 (16)	from 76.2 (10)
Logo deboss 0.5	Both sides	104.94 to 180.24	artwork file	y 25.4 to 303.8	the same

MANUFACTURING ROUTE, ONE BLOCK

STEP	OPERATION	TOOLING	SHEET
1	Slit coil to 272.46 wide; decoil, level and feed	Low	H-2
2	Punch every hole, groove, notch and the logo, in register	High	H-2, H-3
3	Roll form the profile; the hem closes in the last passes	High	H-1
4	Cut to length L after forming	Medium	H-2
5	Bend the four corners in a fixture: BR, TR, TL, BL	Medium	H-1
6	Butt weld the ends; grind flush; prime the weld	Medium	H-1
7	Projection weld the nuts and pins from inside	Low	H-0
8	Rivet the pins into the shelf; clinch in the rails, shelf and strap	Low	H-0, H-5
9	Mask pins and M3 threads; clean, pretreat, powder coat; fill logo on corner blocks	Low	H-0
10	Fit the spring latch with its two screws	Low	H-5

1. x runs from the block's left outer face seen from the front; y up from its bottom outer face. On a flat, s = the run's start + the distance from its corner (table above and view A). A count in () is that many centers at the pitch, starting at the one given.
2. d is depth back from the frame's front edge, q is in from the outer face along the flange; w = d + 63.59 on the web and q + 213.82 on the flange. Positions ± 0.1 to the strips punched datum, accumulated ± 0.3 along the full length.

NOTES

1. Strip: 272.46 $\pm$ 0.2/0 wide, slit from coil, ZM310 both sides, burr toward the inside face.
2. Punch every feature in register before roll forming; the J grooves, joint holes and corner notches share one datum per block length. Cut to length after forming.
3. The weld closes the ring at x 254 on the bottom, between two flange holes and clear of vents (sheet H-1).
4. Mass is the bare formed strip, before nuts, pins and coating.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

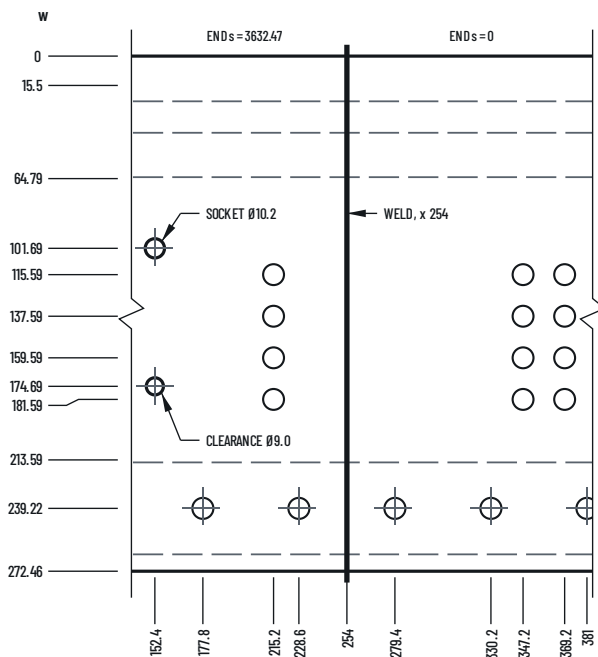
CIRRUS SYSTEMS PROJECT FIREFLY FRAME: HOOP	TITLE HOOP STRIP, FLAT PATTERN Strip lengths, corner positions and every cut feature for the four block sizes			
	PART NO. FF-H01-33 / -32 / -23 / -22		QTY PER BLOCK 1	
 THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	MATERIAL S280GD+ZM310 steel strip, 1.2 [0.047], zinc magnesium coated			
	FINISH Bare at this stage: coated after welding (sheet H-1)			
	SCALE 1:10	UNITS mm	SHEET 11 of 19	DATE 2026-10-05

REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05

H-3

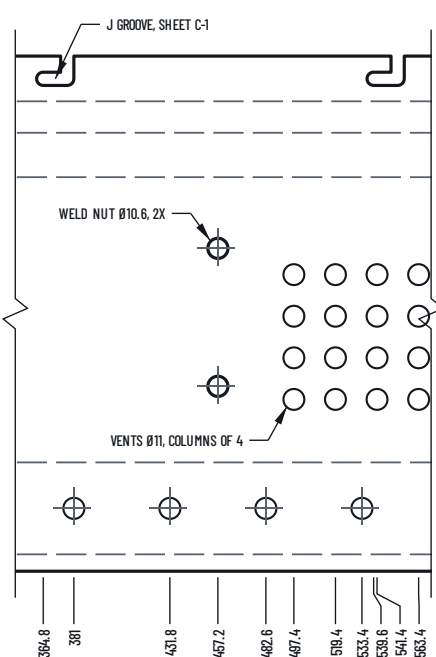


$x \rightarrow .$ Ends butted at the weld



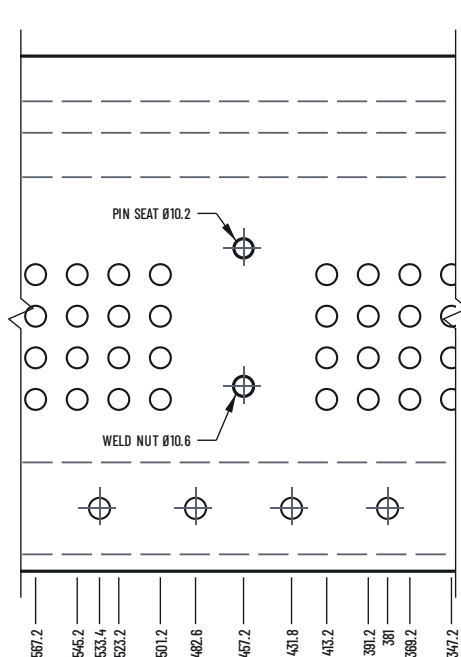
DETAIL E, RIGHT SIDE

$y \rightarrow$. At a joint station



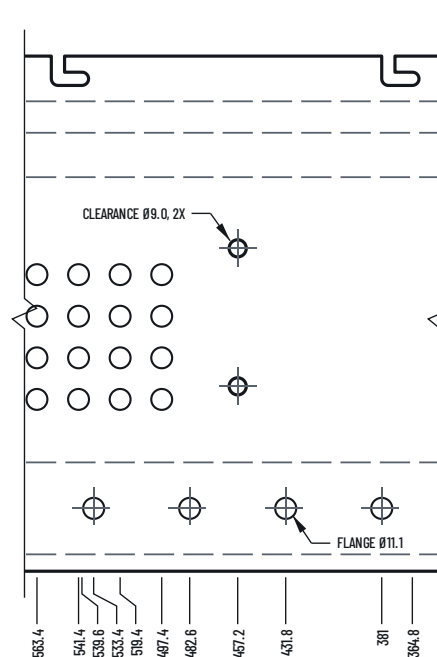
DETAIL G, TOP

← x. At a foot



DETAIL H, LEFT SIDE

← γ . At a joint station



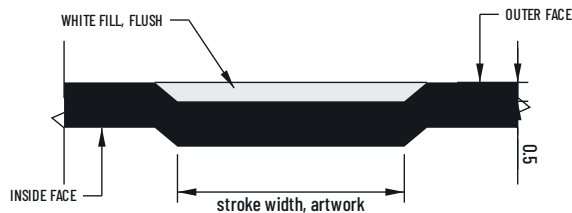
All four windows are on FF-H01-33, at 1:4 and at one height, so every row reads straight across. Positions under each window are block x or y.

PUNCH TOOLS AND HITS, ONE FF-H01-33 STRIP

TOOL	SHAPE AND SIZE	MAKES	HITS	TOOLING
P1	Round Ø9.0	Bolt clearance, bottom and left	7	Low
P2	Round Ø10.2 +0.1/0	Pin sockets and pin seats	6	Low
P3	Round Ø10.6 +0.1/0	Weld nut pilots, top and right	7	Low
P4	Gang of 4, Ø11 at 22	One vent column per hit	68	Medium
P5	Round Ø11.1 [7/16]	Back flange holes	64	Low
P6	J groove, 7 wide, R3.5	Near leg, both sides	12	Medium
P7	Front notch 35.72 x 64.79	U, ply and hem out of each corner	4	Medium
P8	V notch 90°, 11 at the lip	Flange and lip mitre at each corner	4	Medium
P9	Deboss 0.5, logo artwork	Logo, both side webs	2	Medium
P10	Cutoff, full profile	Length L, after forming	1	High

SECTION K-K, LOGO DEBOSS

SCALE 5:1. One stroke of the artwork

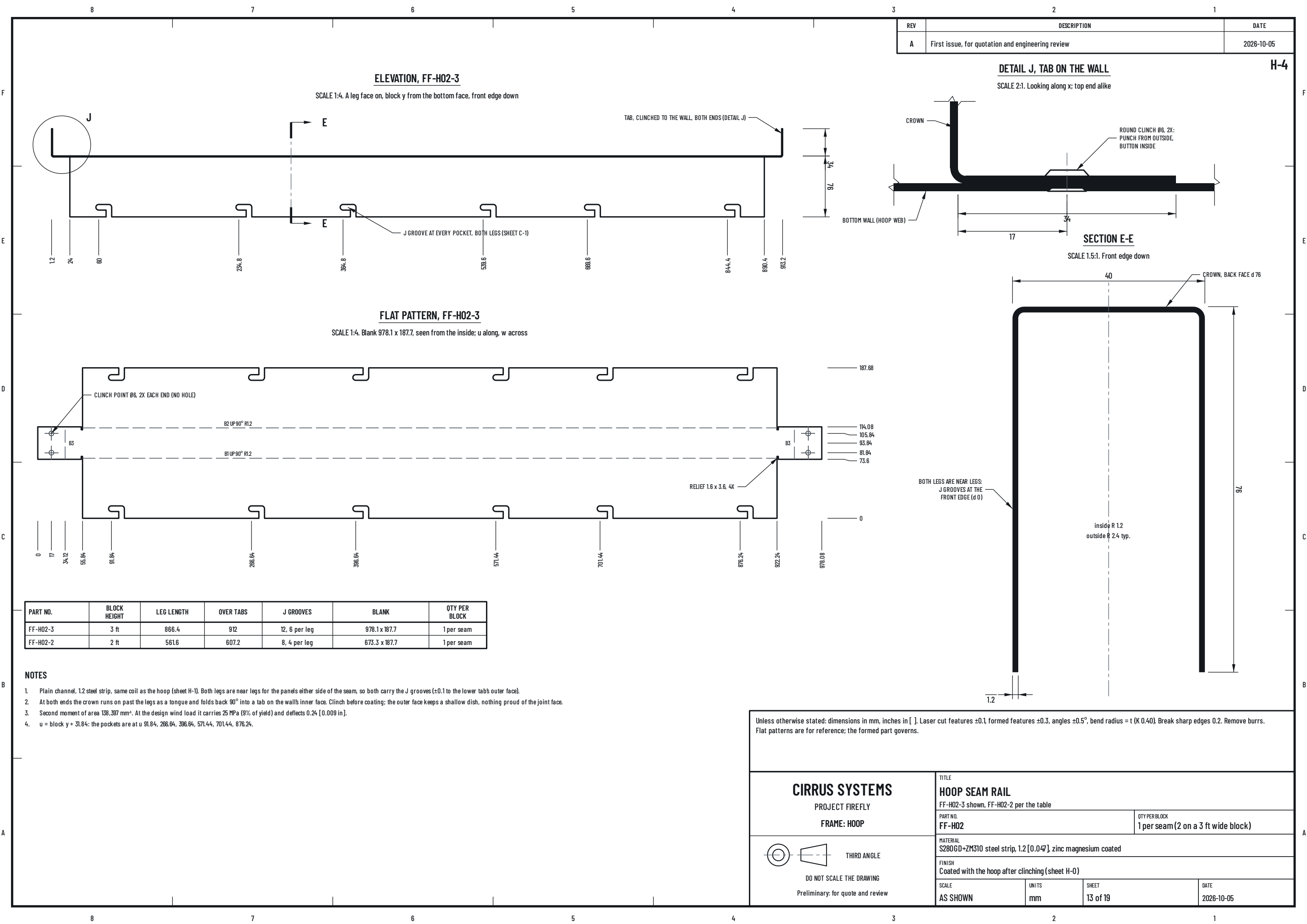


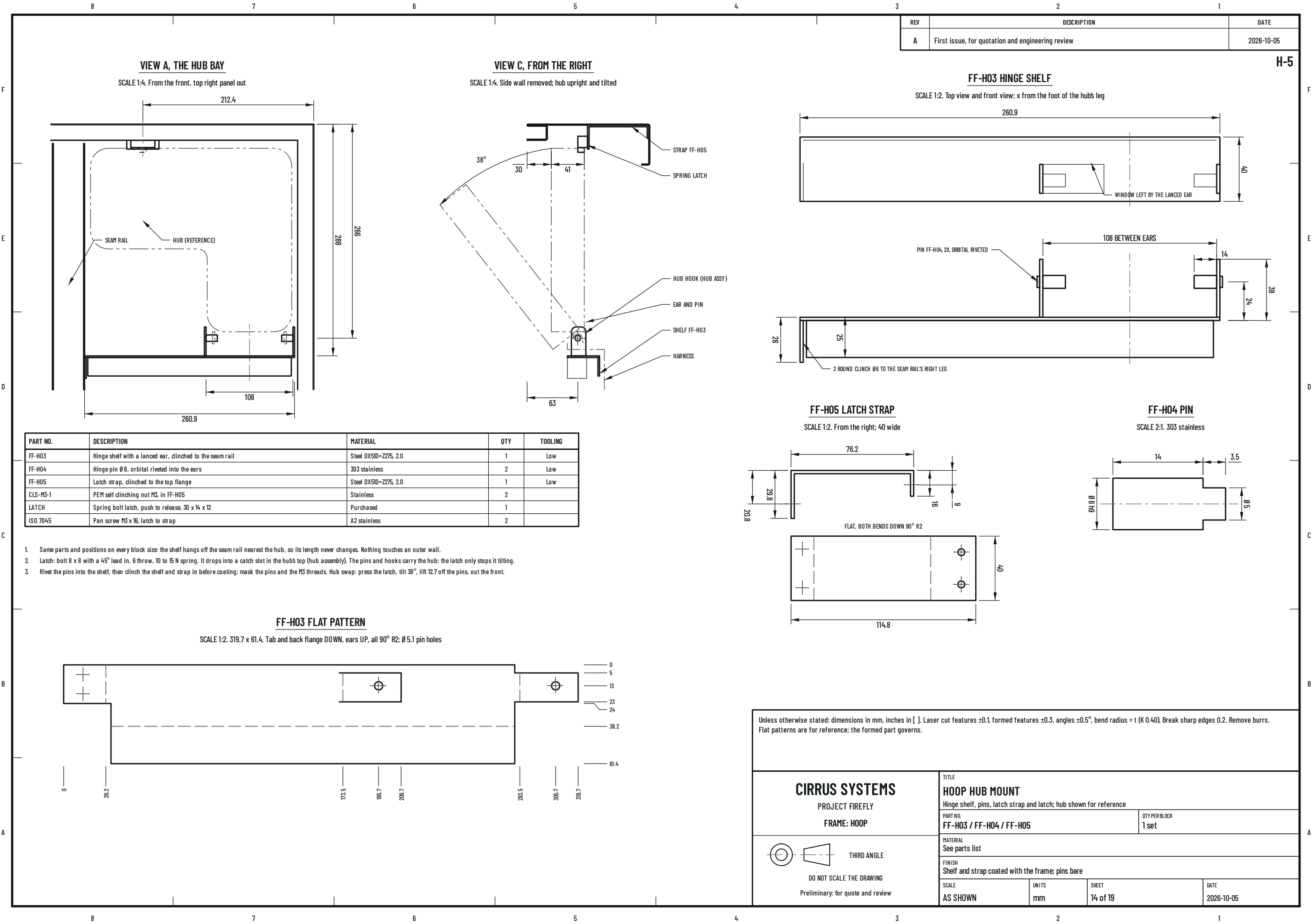
NOTES

1. Punch with the burr toward the inside face. Joint holes Ø9.0 to Ø10.6 true position 0.2 to the punched datum of the block length.
2. Vent columns never fall within 36 of a foot center, 20 of a seam rail or 15 of the weld; the patterns repeat each foot so vents line up block to block.
3. Logo: Cirrus artwork file, 278.4 long and 85.3 high, reading up the side; one tool serves both hands (the left side is the same tool turned end for end).
4. Weld nut pilots and pin seats are sized for the parts on sheet C-5.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS PROJECT FIREFLY FRAME: HOOP	TITLE HOOP STRIP, PUNCH DETAILS Windows on FF-H01-33; every size uses the same tools			
	PART NO. FF-H01-33 / -32 / -23 / -22		QTY PER BLOCK 1	
  THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	MATERIAL S280GD+ZM310 steel strip, 1.2 [0.047], zinc magnesium coated			
	FINISH Bare at this stage: coated after welding (sheet H-1)			
	SCALE AS SHOWN	UNITS mm	SHEET 12 of 19	DATE 2026-10-05

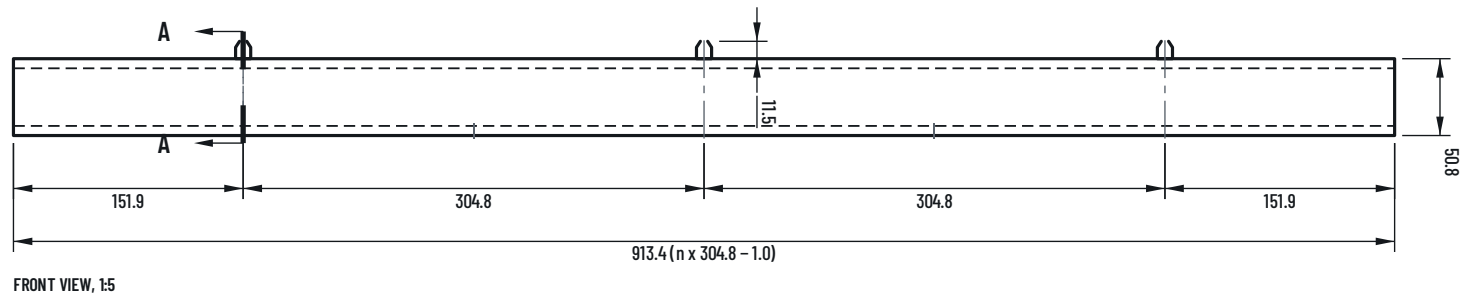
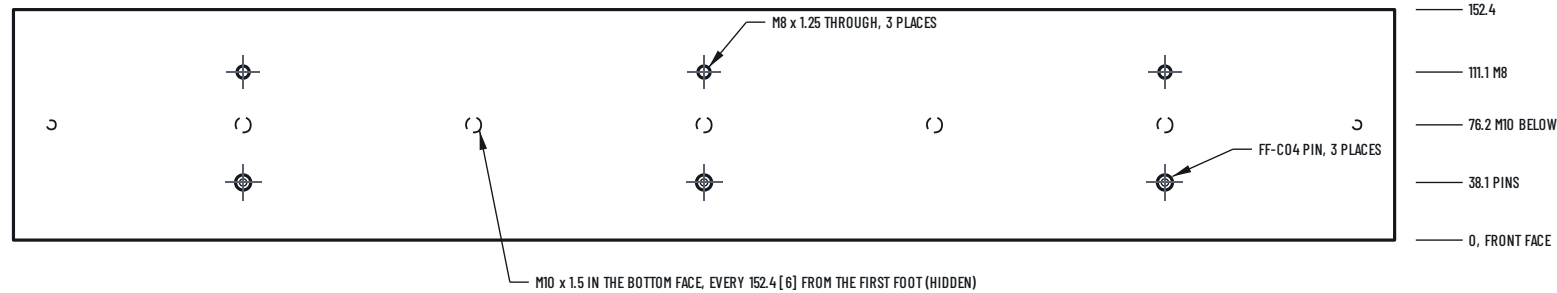




REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05

FF-C03-3 BASE RAIL, 3 FT SECTION

SCALE 1:5. Top view, front face at the bottom; other lengths per the table



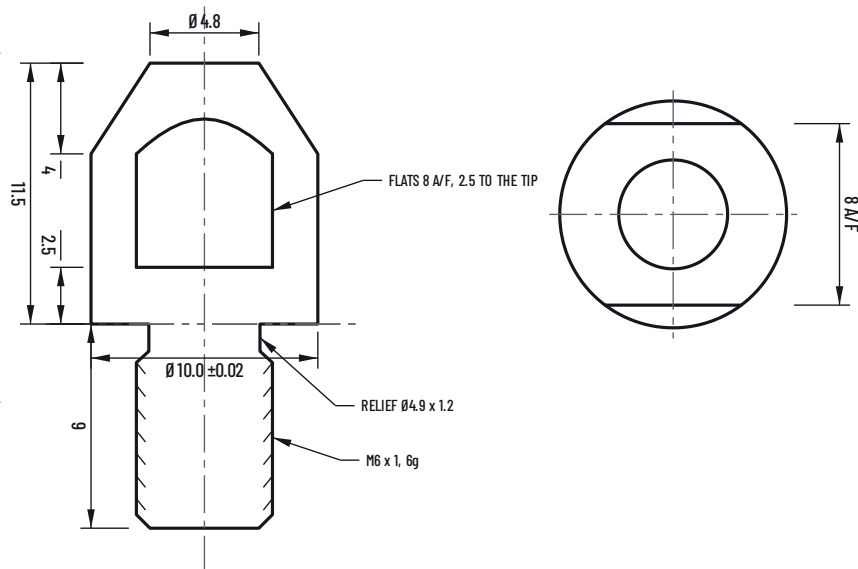
RAIL SECTIONS

PART NO.	FEET	LENGTH	FF-C04 PINS	M8 HOLES	M10 HOLES	kg
FF-C03-2	2	608.6	2	2	3	4
FF-C03-3	3	913.4	3	3	5	6
FF-C03-4	4	1218.2	4	4	7	8
FF-C03-5	5	1523	5	5	9	10
FF-C03-6	6	1827.8	6	6	11	12
FF-C03-7	7	2132.6	7	7	13	14
FF-C03-8	8	2437.4	8	8	15	16
FF-C03-9	9	2742.2	9	9	17	18
FF-C03-10	10	3047	10	10	19	20

1. One section for walls up to 10 ft wide. Wider walls take more sections, meeting on a foot line: the first pin of the next section sits 304.8 mm from the last pin of the one before, set with a block hung across the joint or with a gauge. The ends meet with about 1 mm between them.
2. Every Firefly wall width is a whole number of feet, so a set of sections always adds up to the wall.

FF-C04 RAIL PIN, THREADED

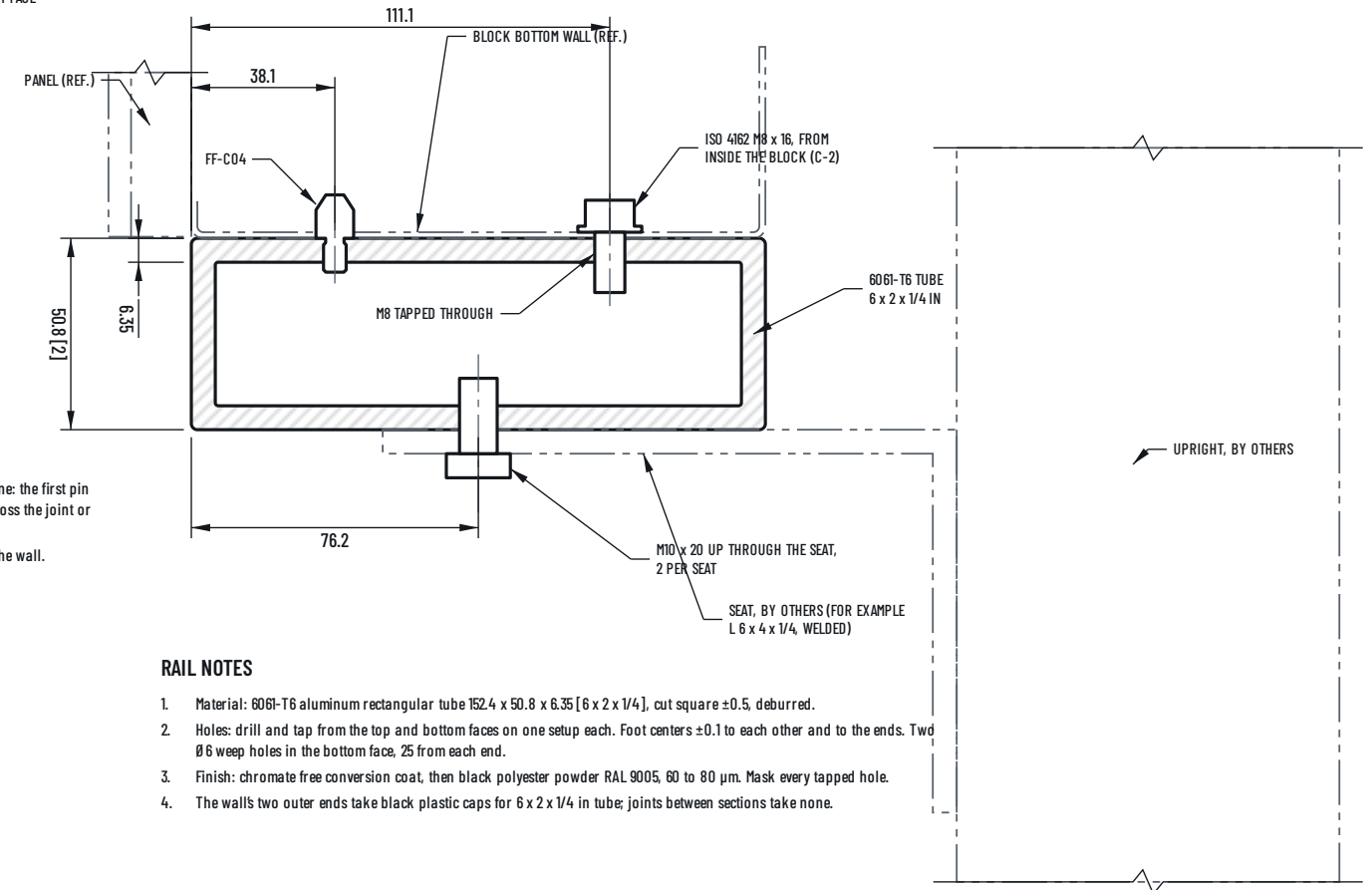
SCALE 3:1. 303 stainless, passivated. Side view with the flats facing; view from above



1. The part above the shoulder matches FF-C01 above a block's top wall: Ø10, 11.5 tall, tapered from 7.5. Only the lowest 2 mm locates in a socket, so the flats cost nothing.
2. Fit after powder coat: medium strength threadlocker, 6 N·m, shoulder down flat on the rail.
3. Ships fitted to the rail.

SECTION A-A, AT A FOOT CENTER

SCALE 1:2. Front face at the left; a block above, the seat behind



RAIL NOTES

1. Material: 6061-T6 aluminum rectangular tube 152.4 x 50.8 x 6.35 [6 x 2 x 1/4], cut square ± 0.5 , deburred.
2. Holes: drill and tap from the top and bottom faces on one setup each. Foot centers ± 0.1 to each other and to the ends. Two $\emptyset 6$ deep holes in the bottom face, 25 from each end.
3. Finish: chromate free conversion coat, then black polyester powder RAL 9005, 60 to 80 μ m. Mask every tapped hole.
4. The walls two outer ends take black plastic caps for 6 x 2 x 1/4 in tube; joints between sections take none.

HOW IT GOES IN

1. Weld or bolt a seat to every upright and add seats between so none are more than 1.52 m [5 ft] apart. Set them level to ± 1 mm across the wall and shim where needed. Every section sits on at least two seats.
2. Lay the rail on the seats, front face in line with the face of the frames above, and bolt it up from below with two M10 x 20 per seat.
3. Hang the first row: drop each block onto the rail pins and run its M8s down into the rail. Tapped aluminum: tighten them to no more than 12 N-m, to be confirmed by test with the C-2 joint torque.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS PROJECT FIREFLY COMMON: RIB AND HOOP	TITLE			
	BASE RAIL			
	Starts the first row: the same pins and bolts as the top of a block			
	PART NO. FF-C03-n, FF-C04		QTY PER BLOCK One rail per wall	
 THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	MATERIAL 6061-T6 tube 6 x 2 x 1/4 in; FF-C04: 303 stainless			
	FINISH Black powder RAL 9005 over conversion coat			
	SCALE AS SHOWN	UNITS mm	SHEET 17 of 19	DATE 2026-10-05

